

3. Chipsets

The chipsets (Platform Controller Hub) and the processor go hand in hand. Certain features of the processor like overclocking and video encoding are only available when a certain chipset is used, so you need to get the combination right or else you are basically throwing money away. So which chipset should you get? Simply Google the model number of your processor and it's respective product page will mention all it's features and list the compatible chipsets. Local vendors will sell a CPU and a motherboard as a unit to avoid compatibility issues so you needn't worry about not getting it right, however, we always advise caution.

4. Overclocking

The process of getting more performance out of your processor is called as overclocking. You do this either by increasing the base clock frequency or by increasing the multiplier. The speed that you see when you buy a processor is the base frequency times the multiplier. So a 3.2 GHz processor may have a base clock of 100 MHz and a multiplier set at 32. Now more processors have their multiplier locked to a very narrow range (+/- 2), when the need for extra power arises this multiplier is raised automatically. Manufacturers have named this technology as Turbo boost, though different manufacturers have different names they essentially do the same thing.

High end processors tend to have an unlocked multiplier, i.e. the multiplier range is wider now, you can scale it up by 16 levels. However, high processing speeds demand a lot more power and that too well regulated power. Also the heat released is tremendous, so the processors are rarely stable at such high speeds. You need to make use of liquid nitrogen and liquid helium to keep temperatures low and ensure a stable performance.

However, the increments obtained are in steps when only the multiplier is fiddled with, getting the teeny bit extra now comes down to the base frequency. A mere 1 MHz increment scales linearly and adds up depending on the current multiplier level. So 1 Mhz base clock increment translates to 32 MHz when the multiplier is 32. Enthusiast overclockers have to achieve the balance between the multiplier and base clock to achieve the maximum possible from your processor. Even then there are external factors like the batch of manufacturing which affects how much a processor can scale.